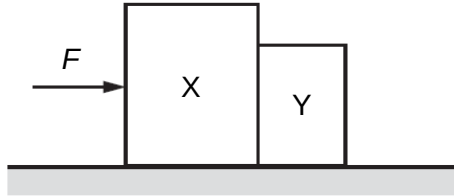


- 8 A single horizontal force F is applied to a block X which is in contact with a separate block Y, as shown.



The blocks remain in contact as they accelerate along a horizontal frictionless surface. X has a greater mass than Y.

Which statement is correct?

- A The acceleration of X is equal to force F divided by the mass of X.
- B The force that X exerts on Y is equal to F .
- C The force that X exerts on Y is less than F .
- D The force that X exerts on Y is less than the force that Y exerts on X.

Answer: C

A: wrong. Acceleration of X = $(F - F_{\text{by Y on X}})$ divided by mass of X

B: wrong. $F_{\text{by X on Y}}$ is equal in magnitude with $F_{\text{by Y on X}}$. If $F_{\text{by Y on X}} = F$, net force on X = 0 and X will not accelerate.

C: correct. $F_{\text{by X on Y}}$ is equal in magnitude with $F_{\text{by Y on X}}$. If $F_{\text{by Y on X}}$ is less than F , net force on X acts in the direction of F and X will accelerate.

D: wrong. By N3L, $F_{\text{by X on Y}}$ is equal in magnitude with $F_{\text{by Y on X}}$.

9 A full barrel of weight W is being rolled up a ramp. The force P is required to hold the barrel at